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Tech Bytes | Asia Pacific

Memory – A Healthy Reset

Corrections are necessary for the cycle to continue. DRAM is the main bottleneck to the AI build-out but the market concerns on the durability of this cycle. We think multiple expansion can be a bigger driver of returns from here. We revise up bear cases substantially for Samsung and SK hynix.

This reset on price performance does not mean the cycle is over. Our core view is that the cycle is still accelerating, earnings revisions remain robust and more sustainable than most believe. The earnings revisions justify the rally – the earnings estimates for memory companies are up just as much as the stocks are – and the LTAs justify a re-rating potential.

Memory cycles – a long-term perspective. We should be near peak cycle by year-end if we go by the historical DRAM cycle duration and a re-rating needs: (1) the supply side to behave, and (2) demand must convert into durable economics in the long-run as well. Prior cycles were never about demand but supply discipline which never held. AI structural demand on top of undisciplined supply is just a longer cycle with a much higher peak, which is a consensus view at the moment.

Memory prices will decline, the cycle always turns as current investment plans come online from late 2027, but price elasticity of AI demand may be different. Lower DRAM prices reduce the cost of running AI inference, making AI deployment cheaper, and create new demand, rather than simply reducing costs for a fixed number of consumer devices.

How much upside? Memory pricing has nearly doubled since February and lead times have stretched out significantly. The underlying driver is constrained physical production due to LTAs. If LTAs could account for 70%+ of total supply in the next 3-5 years, in our view, this implies DRAM stocks could re-rate to at least 8-10x PE vs. 5x PE now, in theory. This is based on simple math of 70%+ of 2027 EPS valued closer to market multiples (rolling 5Y 10-14x PE) and the remaining 30% of non-LTA EPS at historical 5x peak PE of peak earning.

Stock implications. We estimate more than 20-30% DRAM price hikes in 3Q26, enough to keep the YoY rate of change accelerating. Pricing power is translating into earnings revisions that in turn support P/E stability – at 5.2x 2027e earnings – even as stocks surged 70-134% in the last 2-months, and not a change in the discount rate. We revise up our bear case values for SK hynix and Samsung by 175% and 58%, respectively ([Revising up the bear case](#)).

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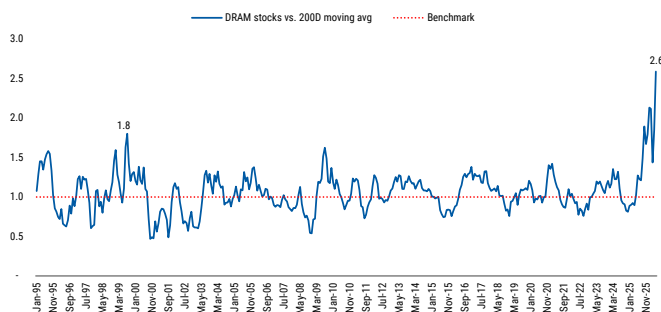
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A Healthy Reset

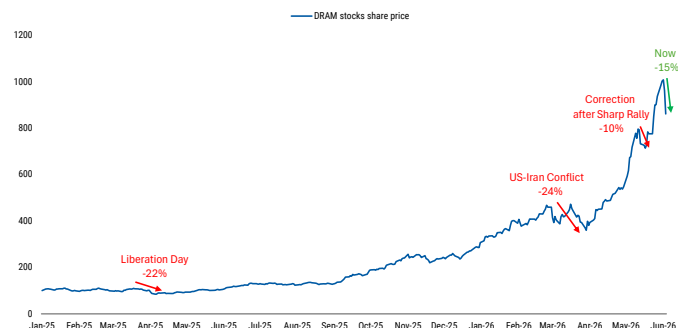
Markets rarely move in a straight line at the pace seen since the March lows. In our view, a correction was inevitable and ultimately healthy if this memory bull market is going to extend into year-end. Parabolic charts in memory stocks clashed with levered ETF exposure and record crowding by hedge funds and retail investors. This reset on price does not mean the cycle is over. Instead, it is more likely that this is a necessary reset for the cycle to eventually extend – we have had 3 such resets since Generative AI was launched in the fall of 2022. Our constructive year-end view is grounded in: (1) still-strong AI demand, (2) supply growth that is limited by clean room and EUV availability, and (3) LTAs driving durable earnings and FCF growth. We are not at peak cycle yet but acknowledge that accelerative moves in fundamentals always reach an apex in terms of the rate of change and stocks tend to react to that.

Exhibit 1: DRAM Stocks vs. 200-Day Moving Average



Source: Factset, Morgan Stanley Research

Exhibit 2: DRAM Stocks: Major sell-offs through the rally



Source: Factset, Morgan Stanley Research

Prior memory corrections after parabolic moves may be instructive. These developments don't necessarily mark the end of a bull market in the memory cycle, but these areas typically don't spring back right away either. That process takes time and this reminds us of similar price moves we have seen in other commodity areas earlier this year (Gold/Silver, Rare Earths, other Industrial Metals, Energy, etc). We would point out that commodities often exhibit this type of volatility while still in the midst of a long bull market.

We note that for the Korea market, historically, post-circuit breaker dynamics skew constructive, with ~75% next-day hit rates improving to ~87-88% over medium horizons, suggesting that while volatility may persist near term, risk-reward is gradually improving into weakness, particularly with Samsung and SK Hynix now trading at ~5.2x and ~4.7x fwd P/E, respectively.

Exhibit 3: Samsung: Returns following KOSPI Circuit Breaker Events

Event	1st trading day	1D	1W	1M	3M
Dotcom Bubble	17/04/2000	9.4%	9.6%	12.0%	36.8%
Dotcom Bubble	18/09/2000	4.8%	-4.3%	-1.8%	-12.1%
Dotcom Bubble	12/09/2001	6.2%	1.5%	-13.3%	60.1%
Covid-19	13/03/2020	0.0%	-8.7%	-6.3%	8.7%
Covid-19	19/03/2020	5.7%	9.3%	13.2%	21.5%
Yen Carry Unwind Scare	06/08/2024	3.0%	3.0%	5.0%	-19.0%
US-Iran Conflict	04/03/2026	11.3%	0.8%	10.2%	109.3%
US-Iran Conflict	09/03/2026	8.3%	5.8%	3.6%	89.6%
Hit Ratio %		87.5%	75.0%	62.5%	75.0%
Mean Return %		6.1%	2.1%	2.8%	36.9%
Max Return %		11.3%	9.6%	13.2%	109.3%
Min Return %		0.0%	-8.7%	-13.3%	-19.0%
Median Return %		5.9%	2.3%	4.3%	29.2%

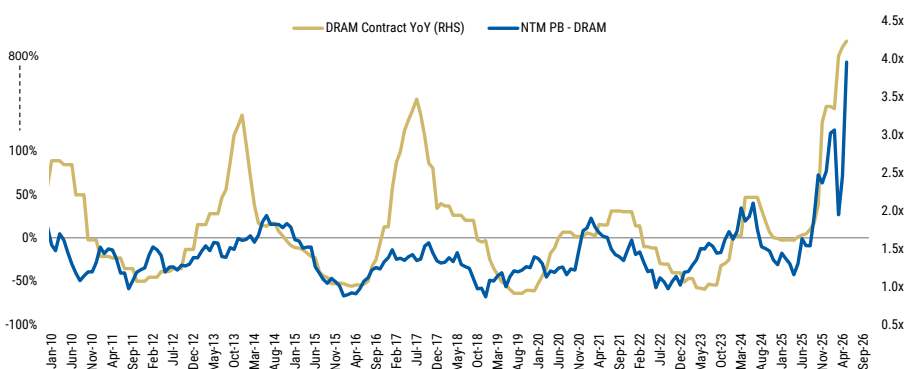
Source: Factset, Morgan Stanley Research

Memory cycles – a long-term perspective

We should be near peak cycle by year-end if we go by the historical DRAM cycle

duration – the playbook has been the typical 6 quarters on the way up before supply becomes less tight (i.e. the YoY rate of change in pricing) and stocks moving about 4-months before that. But because of AI agent demand (which dates from January 2026), it is reasonable to assume the top is at least several quarters away from today. The re-rating needs (1) the supply side to behave, and (2) demand must convert into durable economics in the long-run as well.

- **Prior cycles were never about demand** (every prior cycle had real, growing demand with the exceptions of the 2008 GFC and 2021 initial COVID demand shocks). But the cycles were always supply-driven, racing to meet any demand gap and overshooting.
- **What breaks the cycle is supply discipline**, and that is the variable that has never held. Structural demand on top of undisciplined supply is just a longer cycle with a much higher peak, which is consensus thinking at the moment.

Exhibit 4: Long-Term DRAM cycle YoY chart

Source: Trendforce, Factset, Morgan Stanley Research

What's different? Intelligence vs. consumer electronics demand. Memory has become an essential input for intelligence or token manufacturing where more HBM bandwidth,

more DRAM and NAND capacity mean more intelligence output per second. The larger the model usage, the longer the context and number of users simultaneously supporting more AI agents deployed – this demand is not the same as traditional PC or smartphone where the TAM is fixed.

- AI inference growing exponentially as well as AI agent usage in its infancy and very far from saturation.
- The ratio of memory that goes into PCs and smartphones and other personal-use devices is lower each day as memory capacity keeps moving to AI-related infrastructure.
- Every generation of GPU is constrained by memory, not compute. DRAM content per AI unit is growing 4-7x (from 80GB per A100 to 288-768GB for Rubin GPU/Superchip) while AI chips are growing at about 60% annual run rate today.

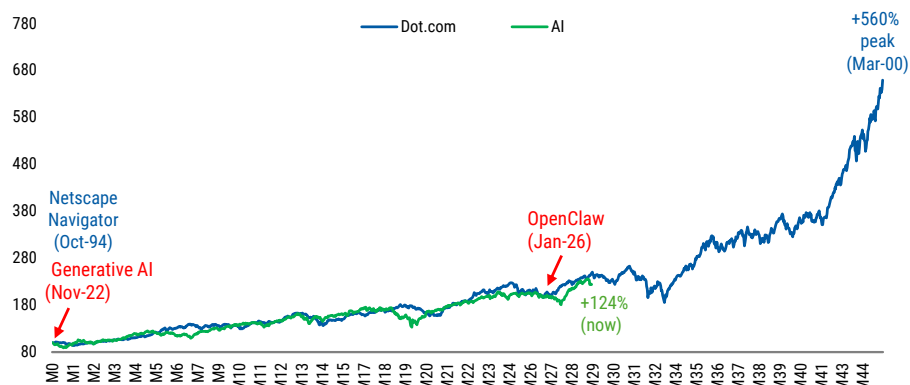
Memory prices will decline, the cycle always turns... as new fab capacity from current investment plans comes online in late 2027. The nature of demand for memory however, is AI-driven and tied to long-lived infrastructure programs rather than short consumer cycles. This gives memory producers more visibility and confidence on future demand but not enough to aggressively overbuild as in the past cycles. According to management targets, adding capacity may be coming in much more controlled stages.

... but price elasticity of AI demand may be different. Lower DRAM prices reduce the cost of running AI inference, making AI deployment cheaper. This creates new demand and expands the addressable market, rather than simply reducing costs for a fixed number of consumer devices.

On bubbles

A bubble can inflate for a long time. We had two great years of stock performance back during that internet build out, and we think we can have at least one more great year this year. If we go back to the internet build out, the Generative AI equivalent of internet – Netscape Navigator – was out at the end of 1994, NASDAQ over the next 4-years was up 149%, and in year-5 in 1999 up 86%. Generative AI came out at the end of 2022, and we are 3-years into this, NASDAQ is up 122%. The Dot.com peak in March 2000 can be explained by: (1) fundamentals where the rate of change in internet traffic growth slowed from doubling every 3-months to doubling every year – that slowdown was not what was built into valuations, and (2) post Fed easy money before Y2K (M2 up >10%), oil prices spiked and the Fed started to withdraw that liquidity. NASDAQ then fell 85% and mean reverted back to mid-1996.

It is hard to predict exactly how much of the current demand bubble will persist in the long-run. Agentic AI is a step change in token generation, up from 20% growth 2-months before OpenClaw and over 120% after its launch on January 2026. Clearly, the current market is distorted by short-term dynamics but which part is base demand and which distortion?

Exhibit 5: AI vs. Dot.com: NASDAQ Performance from the Technology's arrival

Source: Factset, Morgan Stanley Research

Key risks

The risk for the memory cycle at the current stage is plentiful. Technology will evolve, competition and supply will normalize and so will memory stocks. However, the chances of this happening in the very near-term are low, but worth pointing out as it would be significant enough to derail our positive industry view.

- **A technical breakthrough on efficiency** that would use significantly less memory/HBM, or a change that would bypass memory altogether. We would exclude memory compression as it has been a constant, but acknowledge that the industry always looks for ways around bottlenecks.
- **Falling off the AI race.** We still need five major drivers of frontier LLMs and hyperscale AI infrastructure spend to stay elevated.
- **AI demand rate of change** moving off-course and no longer exponential. The ARR (Annualized Run Rate) trajectory is an important indicator of AI usage in enterprise API contracts and the adoption of agentic coding tools, and an important indicator of how the business may grow in the future.
- **Chipflation and the macro impact.** The question is not whether memory prices have risen enough to matter. They have. It is whether the cost is travelling through channels diffused and slow enough that, by the time it becomes fully visible, the underlying price shock will be two years old. [Global Technology: Chipflation – Navigating A Memory Crisis \(2 Jun 2026\)](#).
- **Market liquidity may act as a near-term risk.** The liquidity picture is tightening again from a rate of change standpoint, while China has been aggressively slowing its money supply growth as well. Add in higher oil prices and a very resilient US dollar, and the liquidity picture is less accommodative than it was in the first quarter and a risk given liquidity tends to lead equity markets by approximately 3 months.

The AI CPU memory bull case comes into view

The agentic AI CPU thesis has moved from forecast to validation post GTC Taipei and

Computex 2026. Recent management commentaries are now skewing our server CPU and memory TAM towards the bull case. The cluster CPU-to-GPU ratio that underpinned our previous framework (*Agentic AI - The Surge Begins*) is being corroborated, with commentary across major chip companies pointing to a shift from training-era ratios towards parity for agentic AI workloads, alongside a step-change in per-agent token intensity. This supports the upper end of our US\$283bn bull case server CPU TAM.

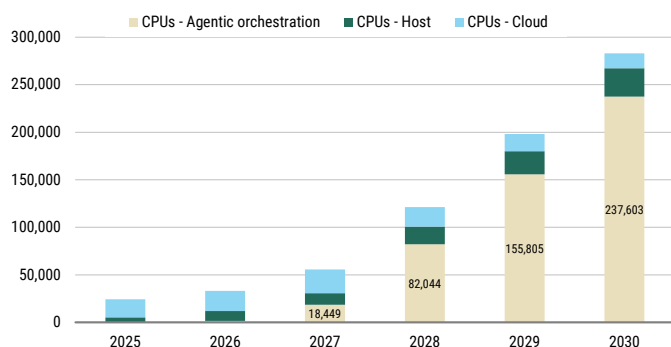
We see the read-through to memory as asymmetric to the upside. The move towards native LPDDR (mobile DRAM) across the latest generation of orchestration CPUs, combined with higher token intensity per agent, supports greater DRAM content per CPU – the key swing factor in our framework. We raise our incremental DRAM estimate accordingly (prior ~74EB base case/ ~221EB bull case).

Significant incremental upside – when intelligence is produced at the point of context.

The data center is coming to laptops, smartphones and at home. So wherever and whenever the context is, that is where intelligence will be produced – where we see the upside will likely be coming from edge device, such as agentic AI PCs introduced at Computex that come with a memory vector that server-attached framework excludes today. With 128GB of unified LPDDR5x per AI PC systems, we see this as a source of potential upside to both our memory TAM.

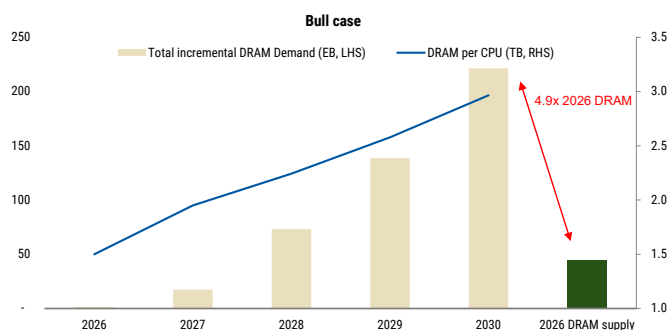
- **Agents will be on premise because that is where all of the data sits** (secure, proprietary and all the skills associated with users/companies).
- **Agents iterate until it can get the job done using all different tools.** The amount of computation that is necessary... because it is running autonomously for so long, and instead of responding to a query, the amount on computation has grown exponentially.
- **Instead of humans using tools, it is agents using tools** – people use tools every now and then, agents use tools all the time and very quickly. You give digital agents a PC and access to GPU and memory.

Exhibit 6: We estimate the Agentic CPU TAM to grow at a 251% CAGR from FY26 to FY30



Source: Morgan Stanley Research estimates

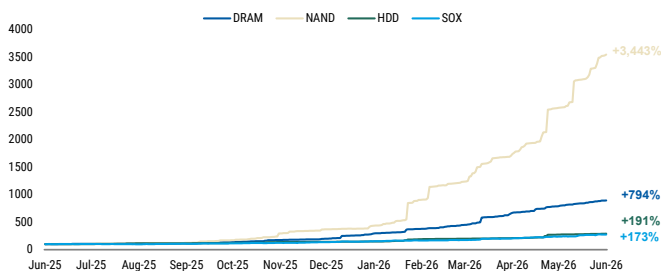
Exhibit 7: We estimate incremental DRAM demand to reach ~221EB as orchestration CPU racks scale



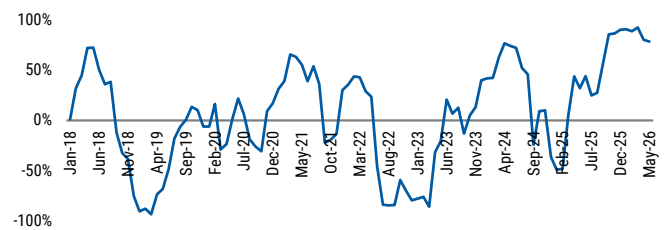
Source: Morgan Stanley Research estimates

What have companies said?

- **NVIDIA (covered by Joseph Moore):** CEO Jensen Huang said: **"Every CPU today was designed for humans. Vera was designed for agents."** Vera Rubin is in full production, with a supply chain twice the size of Grace Blackwell; Vera is pitched as a purpose-built CPU for agentic and reinforcement learning workloads, claiming ~2x efficiency and 50% faster performance vs traditional x86 server CPUs. Huang stressed single-thread performance over core count and suggested Vera can sit across GPU head nodes, storage servers and orchestration nodes.
- **Intel (covered by Joseph Moore):** CEO Lip-Bu Tan said **the CPU:GPU ratio has shifted from 1:8 in frontier-model training to 1:1 or better for agentic AI**, as the CPU orchestrates the reasoning process, while an agent can consume up to ~1,000x more tokens than single-turn reasoning. Intel launched the Xeon 6+ (Clearwater Forest, Intel 18A, 288 e-cores, 576MB L3), claiming up to 150,000 agents per rack, previewed rackscale "Intelligence Center" designs with Foxconn, and confirmed big-core Diamond Rapids slips to 2027.
- **Arm (covered by Lee Simpson):** Arm holds ~50% of CPU compute among top hyperscalers, with AGI CPU customer demand now >US\$2bn across FYE27-28 (more than double the level indicated at Arm Everywhere), and the AGI CPU co-developed with Meta for agentic infrastructure. Arm unveiled two AGI-CPU rack reference designs – a 36kW air-cooled rack with 8,160 cores and a 200kW liquid-cooled rack with 45,696 cores – with data-centre royalties more than doubling YoY.
- **Qualcomm (covered by Joseph Moore):** CEO Cristiano Amon declared 2026 the "Year of the Agent," with agentic AI moving beyond prompts to take action across phones, PCs, vehicles, robots, industrial systems and cloud. Qualcomm announced Dragonfly, its official entry into the high-performance data centre server market, alongside a new Snapdragon C platform for Windows laptops and the Dragonwing IQ10 robotics reference design.

Exhibit 8: NTM consensus EPS trend over last 1 year

Source: Factset, Morgan Stanley Research

Exhibit 9: Memory's earnings revision breadth since 2018

Source: Factset, Morgan Stanley Research

Memory Channel Checks

DRAM

- SemiAnalysis suggests a reduction in the Rubin NVL72 platform's SOCAMM memory configuration from 55TB to 28TB per rack ([news](#)) The spec downgrade is consistent with our channel checks. However, our channel checks also indicate that the overall Nvidia memory order size does not shrink, which indicates the DRAM shortage has not come to the double order stage yet. (See more comments from Joe Moore: [Weekly: NVDA memory cuts? And Apr SIA](#))
- HBM pricing negotiations are still ongoing and prices will likely be raised by at least 50-100% YoY starting from 2027. The 2026 HBM pricing has already been settled under annual contract negotiated earlier last year and we do not expect near-term change in 2H26.
- The pricing for 3Q26 will continue to increase and we expect +20% QoQ ask at the very beginning but we've already seen signs of consumer electronics customers starting to cut orders after 2Q26 pricing hikes. Therefore, we expect the 3Q trend to diverge between AI vs non-AI products
- LTA negotiations are still ongoing, amid AI growth and structural shortages that will likely persist into 2028, some of the suppliers started to de-commit hyperscale customers in order to win more flexibility on quarterly pricing negotiation.
- We expect AI related products to continue gain share in the supply allocation from over 50% this year to nearly 70% next year. Besides additional supply from new capacity, the consumer electronics and module makers will likely face continued order squeeze due to tight AI demand growth

NAND

- 3Q26 contract pricing will likely increase by another 30% QoQ mainly driven by eSSDs. Spot pricing trends muted given the consumer electronics market already faces challenges on digesting the soaring pricing.
- HBF products outlook remain mixed according to the industry feedback. CSPs are worried that the durability issue of NAND die will significantly lower the long cycle of GPU when they are packaged together and eSSD seems to be a better/safer solution for now. One customer appears interested in HBF technology for its smartphone and edge devices to be compatible with its AI applications.
- We expect AI NAND to grow with at least 50% YoY bit growth moving into 2027, mainly driven by CSP's KV Cache offloading demand as well as the QLC SSD capacity expansion in response to the shortage in HDD side. The agentic rack could potentially lift the TAM for AI NAND demand as well, as we estimate 2-8TB eSSD buffer per tray while the specific rack shipment will become more meaningful moving into 2027.

Nvidia AI PC

At GTC Taipei, Jensen Huang unveiled the RTX Spark, Nvidia's first SoC for Windows PCs and its entry into consumer PC silicon, against Apple Silicon and Qualcomm Snapdragon X2. Huang framed the PC as moving "from tool to teammate" - you ask, and the PC does the work. The key point is that AI agents become the main interface and the largest users of compute, with much of that work running on-device rather than in the cloud. Local execution keeps sensitive context on the machine, runs persistent background agents, and cuts latency versus a 100%-cloud setup.

For memory, this means more DRAM per device, because running large models on the device needs a big pool of memory. RTX Spark has a 20-core Grace Arm CPU (10x Cortex-X925 + 10x Cortex-A725, co-developed with MediaTek) and a Blackwell GPU (6,144 CUDA cores) with up to 128GB unified LPDDR5X, up to 300GB/s memory bandwidth, on TSMC 3nm. It can run 120bn-parameter models on the device. NAND depends on the OEM and the model- ASUS ProArt offers 2TB SSD on the P16 and 1TB on the P14, with spare slots for expansion, while most other options are not confirmed yet.

On pricing, our [hardware team estimates](#) the highest-end spec notebook will likely exceed US\$3K, but it also comes in a desktop form factor. Preliminarily, their checks suggest [<10M units of cumulative shipments](#) over the next 3 years vs. an annual PC TAM of ~270M units given higher price points.

Memory inflections

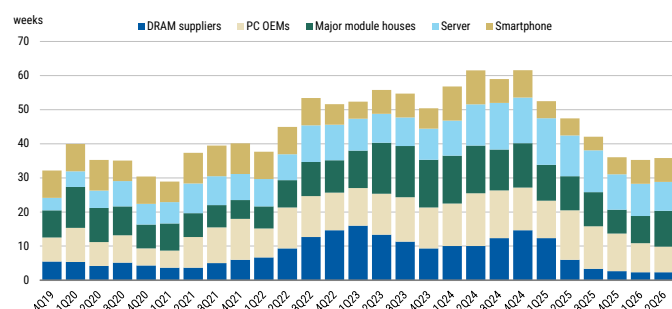
We monitor four signposts as early inflection signals during a cycle:

1. **DRAM prices extending strength.** We expect LTAs to change the high volatility nature of memory pricing for the next 2-3 years. After significant hikes from 4Q25-2Q26, we expect pricing increases on a QoQ basis to narrow to around 8-13% QoQ for both DRAM and NAND. Instead of viewing this as a sign of inflection, we believe LTAs can instill stability, improve earnings predictability, and reduce the risk of overbuilding capacity with better supply management for memory companies by locking in pricing at an elevated level with prepayments. ([LTAs – How They Are Affecting Memory](#))
2. **Inventory adjustment cycle.** Supplier inventory remains at historical lows of only 2-3 weeks for DRAM and 4-5 weeks for NAND at the beginning of 2Q26. Server customers, despite aggressive orders, continue to work down their inventory, with remaining end customers retaining healthy inventory levels. During an upturn, lead times for the delivery of chips to manufacturers of PCs, servers and smartphones tend to grow longer. This prompts those customers to order more than they need and hoard that inventory in case longer lead times grow into a real shortage. We see this as an ongoing issue for consumer customers, who face a trade-off between soaring prices and limited supply allocation.
3. **Capex budgets pared down while capital intensity moves higher.** There is typically either too much or too little supply in memory. The industry is competitive but also rational, and should therefore be responsive to periods of price disequilibrium. In response to the increase in inventories and the prospect of lower profitability, chip manufacturers reduce their capacity growth to manufacture chips to meet that demand. Capital expenditure for DRAM was cut

by 23% in 2019 and stayed flat in 2020, while the 2015 downturn saw only 9% cuts. However, AI demand is likely to keep absolute capex levels elevated in upcoming downturns and also prompt companies to require higher returns to justify those investments.

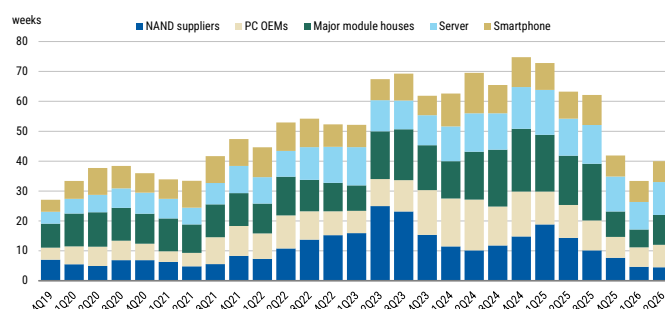
4. **Continued upward earnings revisions and valuation expansion.** The market often runs ahead of the actual revisions in earnings, and we would look for further earnings upside through HBM pricing negotiations. Our sensitivity analysis ([here](#)) suggests that the market is not yet assigning any meaningful P/E premium to memory suppliers' LTA-backed earnings and FCF. For illustration, if we were to apply a 10x P/E multiple to LTA-backed earnings and assume 70% commodity LTA coverage, the 2027 P/E for Samsung/hynix could expand to ~8.5x, all else equal (from 5x currently).

Exhibit 10: DRAM inventory



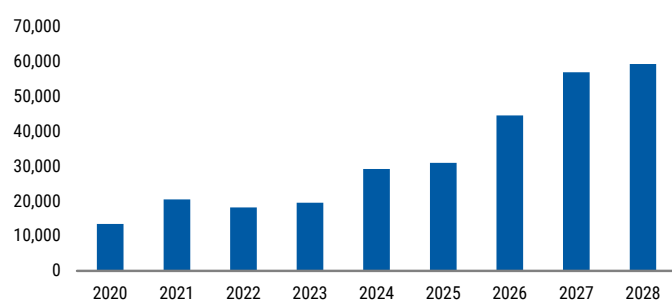
Source: TrendForce estimates, Morgan Stanley Research

Exhibit 11: NAND inventory



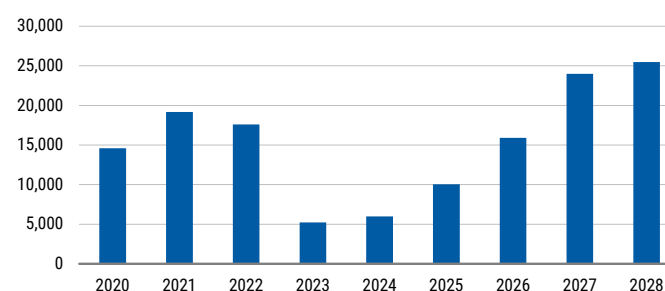
Source: TrendForce estimates, Morgan Stanley Research

Exhibit 12: DRAM Capex (US\$ mn)



Source: TrendForce estimates, Morgan Stanley Research

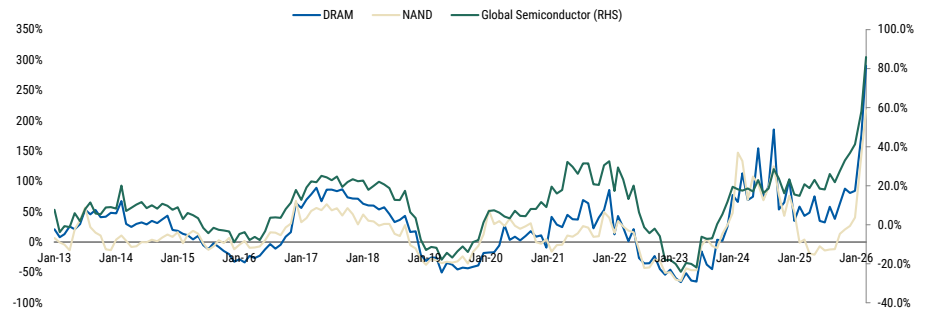
Exhibit 13: NAND Capex (US\$ mn)



Source: TrendForce estimates, Morgan Stanley Research

Exhibit 14:

DRAM in the context of the semiconductor cycle – revenue YoY is highly correlated for memory with DRAM at a more advanced stage relative to NAND



Source: SIA

Revising up the bear case

With the visibility of the persistent shortage in 2027 becoming clear, we revise up our bear case scenarios to assume a downcycle in 2028, instead of going back to a downcycle with 1x P/B in our original assumption. For our bear cases, we assume the commodity memory pricing will face -30-50% YoY decrease due to a pause in AI capex or the industry supply catching up faster than expected. Over the long-term, we assumed that margin for HBM will decrease to ~40% due to market competition, DRAM margin will go back to 20-30% and NAND margin will normalize in the low teens% as in the previous cycle.

As such, our new bear case valuations on residual income models are W11mn for SK hynix, implying 2.4x 27e P/E and W190,000 for Samsung Electronics (005930.KS) and W152,000 for Samsung Electronics preferred shares (005935.KS), implying 3.1x 27e P/E.

Risk Reward – SK hynix (000660.KS)

Robust commodity super-cycle supported by AI in 2026

PRICE TARGET W2,600,000

Base case, residual income valuation model. We assume a cost of equity of 11.5% (5% risk-free rate, 6.5% risk premium, 1.0 beta) and a terminal growth rate of 3%.

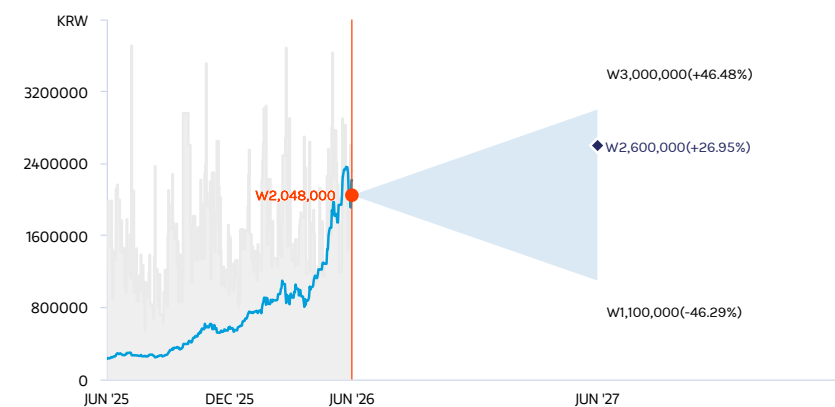
Consensus Price Target Distribution

W680,000.00 W2,531,218 W4,049,273.00

Source: Refinitiv, Morgan Stanley Research



RISK REWARD CHART



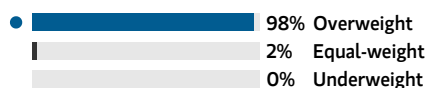
Key: — Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

OVERWEIGHT THESIS

- We believe the commodity cycle is likely to stay robust in 2026 and into 2027, thanks to AI inference demand growth.
- Our earnings estimates reflect downside protection from LTAs into 2028, stronger commodity pricing assumptions, and stronger HBM pricing reset in 2027.
- We do not think this is the time to rotate out of outperformers such as SK hynix.
- Our sensitivity analysis suggests that the market is not yet assigning any meaningful P/E premium to the company's LTA-backed memory earnings and FCF.

Consensus Rating Distribution



● MS Rating

Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Pricing Power: *Positive*
 Secular Growth: *Positive*
 Self-help: *Positive*

View descriptions of Risk Rewards Themes [here](#)

BULL CASE W3,000,000

Memory TAM grows; AI HBM leadership is extended

We see a stronger-than-expected commodity cycle for DRAM and NAND, supported by AI inference demand moving into 2027. Our bull scenario valuation implies 6.7x 2027e P/E.

BASE CASE W2,600,000

Commodity up-cycle

We see a stronger-than-expected commodity cycle for DRAM and NAND, supported by AI inference demand in 2026 and into 2027. HBM competition risks remain but are being increasingly recognized by the market. Our price target implies 5.7x 2027e P/E, underpinned by Hynix's leadership in HBM and a sharp rebound in commodity memory pricing.

BEAR CASE W1,100,000

2028 down cycle

We assume a period of AI computing digestion in 2028, leading to more aggressive margin erosion for the HBM segment owing to competitors' aggressive capacity expansion and pricing competition, as well as faster DRAM pricing drops with muted demand recovery. Our bear case valuation implies 2.4x 2027e P/B.

Risk Reward – SK hynix (000660.KS)

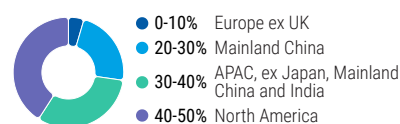
KEY EARNINGS INPUTS

Drivers	2025	2026e	2027e	2028e
Revenue Growth (%)	125.9	679.1	1,472.8	785.3
OPM (%)	48.6	80.0	81.5	78.7
CAPEX (%)	(2,840,890.0)	(6,257,430.9)	(7,730,302.6)	(8,790,428.5)

INVESTMENT DRIVERS

- Supply and demand outlook for DRAM, NAND, and HBM
- Global demand for data center products (server DRAM, enterprise SSD)

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies [here](#)

MS ALPHA MODELS

1/5
MOST

3 Month
Horizon

Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- ASP growth re-acceleration
- Preemptive production cuts or disruption
- Sharp demand improvements
- Significant increase in capital returns

RISKS TO DOWNSIDE

- End demand turning weaker than expected
- Rising competition for DDR5, causing overspending on the supply side.
- Inventories remaining elevated at cloud and Chinese smartphone customers.

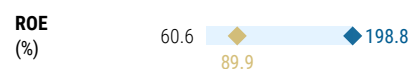
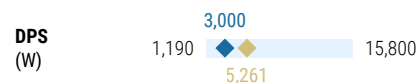
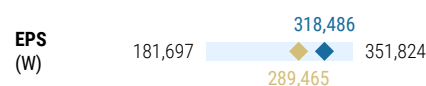
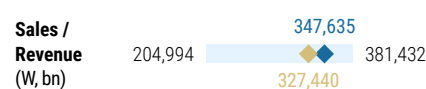
OWNERSHIP POSITIONING

Inst. Owners, % Active 78.6%

Source: Refinitiv, Morgan Stanley Research

MS ESTIMATES VS. CONSENSUS

FY Dec 2026e



◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

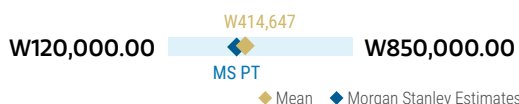
Risk Reward – Samsung Electronics (005930.KS) Top Pick

Improvement in HBM + commodity up-cycle = stock re-rating

PRICE TARGET W381,000

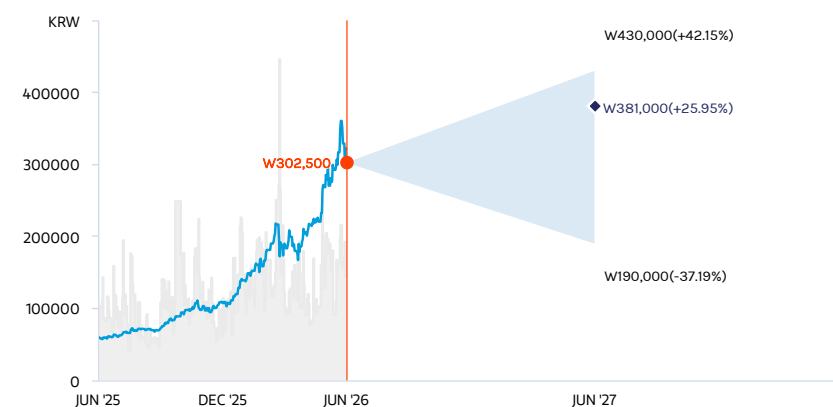
Our base case price target is derived from our residual income valuation model. At our price target, the 2027e P/B multiple would be c.2x, in line with the stock's commodity cycle peak of 2.0x. We assume an 11.5% cost of equity, based on a beta of 1.0, and our terminal growth rate assumption is 3%.

Consensus Price Target Distribution



Source: Refinitiv, Morgan Stanley Research

RISK REWARD CHART



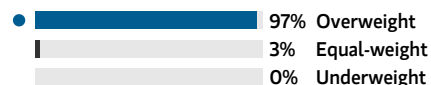
Key: — Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

OVERWEIGHT THESIS

- We expect pricing to move well beyond historical mean reversion from here, on a price per gigabit basis, which should create further incremental tailwinds to forecasts for 2026-27. The commodity cycle is better than feared thanks to AI inference demand.
- We expect HBM improvement and potential market share gains from 2026.
- Rising prices and falling inventories, along with supply cuts and improving demand, are positives for margin recovery in the memory business.
- The implied 2027e P/B at our price target is 2x, aligning with the stock's commodity cycle peak of 2.0x.

Consensus Rating Distribution



● MS Rating

Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Secular Growth: *Positive*
Self-help: *Positive*

View descriptions of Risk Rewards Themes [here](#)

BULL CASE

W430,000

HPC drives sustained memory pricing

High-performance computing drives sustained memory pricing strength and capital returns exceed expectations: We assume continued robust demand growth for memory and OLED with stabilization in the macro environment and China's failure to successfully enter the memory semiconductor industry. Our bull case valuation is based 14x HBM, 5x conventional memory and 10x P/E for the rest of the group profitability.

BASE CASE

W381,000

Commodity cycle recovery

We expect favorable commodity pricing trends to bode well for Samsung's share price with improving HBM market share and new product execution. Our base case reflects our relatively conservative view on HBM progress this year, but sequential improvement in 2026 and beyond support Samsung's earnings growth and stock re-rating amid low expectations. 2027e implied P/B at c.2x.

BEAR CASE

W190,000

Slower memory consumption, competition from China

We assume a period of AI computing digestion in 2028, leading to more aggressive margin erosion for the HBM segment owing to competitors' aggressive capacity expansion and pricing competition, as well as faster DRAM pricing drops with muted demand recovery. Our bear case valuation implies 3.1x 2027e P/B.

Risk Reward – Samsung Electronics (005930.KS)

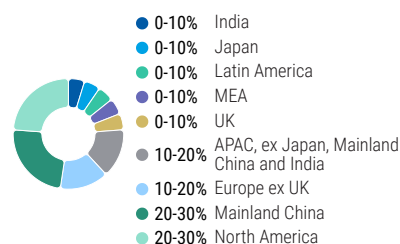
KEY EARNINGS INPUTS

Drivers	2025	2026e	2027e	2028e
Revenue Growth (%)	10.9	120.8	39.2	6.3
OPM (%)	13.1	57.3	62.5	58.7
CAPEX (%)	(4,730,271.0)	(14,538,562.5)	(13,759,464.7)	(13,426,163.4)

INVESTMENT DRIVERS

- Supply-demand outlook for memory and OLED
- Smartphone market share and margins
- Capital returns of 50% FCF

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies [here](#)

MS ALPHA MODELS

1/5
MOST

3 Month
Horizon

Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- New technological developments, especially in memory and foldable displays
- Longer-lasting memory up-cycle from AI and hyperscale data center growth

RISKS TO DOWNSIDE

- Product and memory cycle, including Apple and new Chinese smartphone competition
- Earnings growth concentration in semiconductors

OWNERSHIP POSITIONING

Inst. Owners, % Active **84.1%**

Source: Refinitiv, Morgan Stanley Research

MS ESTIMATES VS. CONSENSUS

FY Dec 2026e

Sales / Revenue (W, bn) **736,813**
502,377 **677,605** 802,158

EBITDA (W, bn) **461,966**
134,278 **386,864** 500,419

EBIT (W, bn) **420,719**
77,082 **335,775** 455,878

EPS (W) **53,985**
11,782 **42,575** 56,372

◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

Risk Reward – Samsung Electronics (005935.KS)

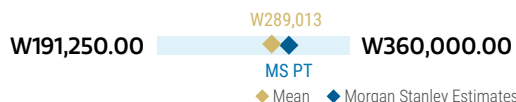
Overweight with discount to common shares to narrow

PRICE TARGET W304,800

Our price target for the preferred shares is based on our PT for the common shares and our assumption that the preferred shares' trade at a 20% discount to the common shares, based on the past 2-year average. We assume an 11.5% cost of equity, based on a beta of 1.0, and our terminal growth rate assumption is 3%.

Consensus Price Target Distribution

Source: Refinitiv, Morgan Stanley Research



RISK REWARD CHART



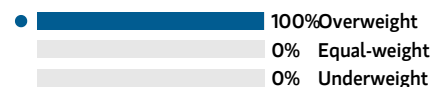
Key: — Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

OVERWEIGHT THESIS

- We expect pricing to move well beyond historical mean reversion from here, on a price per gigabit basis, which should create further incremental tailwinds to forecasts for 2026-27. The commodity cycle is better than feared thanks to AI inference demand.
- We expect HBM improvement and potential market share gains from 2026.
- Rising prices and falling inventories, along with supply cuts and improving demand, are positives for margin recovery in the memory business.
- Common shares' implied 2027e P/E is 5x. We assume that the preferred shares' discount to the common shares is 20% (the past 2-year average).

Consensus Rating Distribution



● MS Rating

Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Secular Growth: *Positive*

Self-help: *Positive*

View descriptions of Risk Rewards Themes [here](#)

BULL CASE

W344,000

20% discount to bull case value for common shares

High-performance computing drives sustained memory pricing strength and capital returns exceed expectations: We assume continued robust demand growth for memory and OLED with stabilization in the macro environment and China's failure to successfully enter the memory semiconductor industry.

BASE CASE

W304,800

20% discount to base case value for common shares

We expect favorable commodity pricing trends to bode well for Samsung's share price with improving HBM market share and new product execution. Our base case reflects our optimistic view on commodity upcycle and stock re-rating with LTA to potentially smooth the cyclical of memory sector.

BEAR CASE

W152,000

20% discount to bear case value for common shares

We assume a period of AI computing digestion in 2028, leading to more aggressive margin erosion for the HBM segment owing to competitors' aggressive capacity expansion and pricing competition, as well as faster DRAM pricing drops with muted demand recovery. Our bear case valuation implies 3.1x 2027e P/B.

Risk Reward – Samsung Electronics (005935.KS)

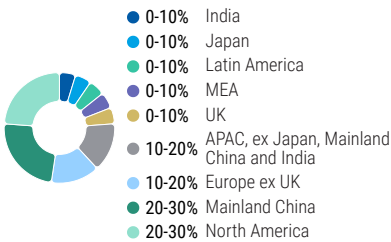
KEY EARNINGS INPUTS

Drivers	2025	2026e	2027e	2028e
Revenue Growth (%)	10.9	120.8	39.2	6.3
OPM (%)	13.1	57.3	62.5	58.7
CAPEX (%)	(4,730,271.0)	(14,538,562.3)	(13,759,464.7)	(13,426,163.4)

INVESTMENT DRIVERS

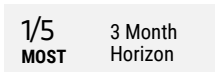
- Supply-demand outlook for memory and OLED
- Smartphone market share and margins
- Capital returns of 50% FCF

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies [here](#)

MS ALPHA MODELS



Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- New technology developments, especially in memory and foldable displays
- Longer-lasting memory up-cycle from AI and hyperscale data center growth
- Further narrowing of discount to common shares

RISKS TO DOWNSIDE

- Product and memory cycle, including Apple and new Chinese smartphone competition
- Earnings growth concentration in semiconductors
- Deepening of discount to common shares

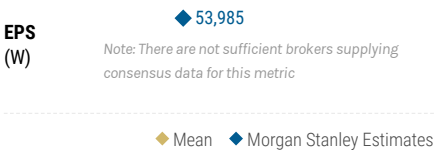
OWNERSHIP POSITIONING



Source: Refinitiv, Morgan Stanley Research

MS ESTIMATES VS. CONSENSUS

FY Dec 2026e



Source: Refinitiv, Morgan Stanley Research

Risk Reward Reference links

1. View explanation of Options Probabilities methodology - [Options_Probabilities_Exhibit_Link.pdf](#)
2. View descriptions of Risk Rewards Themes - [RR_Themes_Exhibit_Link.pdf](#)
3. View explanation of regional hierarchies - [GEG_Exhibit_Link.pdf](#)
4. View explanation of Theme/Exposure methodology - [ESG_Sustainable_Solutions_External_Link.pdf](#)
5. View explanation of HERS methodology - [ESG_HERS_External_Link.pdf](#)

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Analyst Certification

The following analysts hereby certify that their views about the companies and their securities discussed in this report are accurately expressed and that they have not received and will not receive direct or indirect compensation in exchange for expressing specific recommendations or views in this report: Cindy Huang; Ryan Kim; Shawn Kim; Duan Liu.

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Within the last 12 months, Morgan Stanley managed or co-managed a public offering (or 144A offering) of securities of Samsung Electronics, SK hynix.

Within the last 12 months, Morgan Stanley has received compensation for investment banking services from SK hynix.

In the next 3 months, Morgan Stanley expects to receive or intends to seek compensation for investment banking services from Samsung Electronics, SK hynix.

Within the last 12 months, Morgan Stanley has provided or is providing investment banking services to, or has an investment banking client relationship with, the following company: Samsung Electronics, SK hynix.

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Certain disclosures listed above are also for compliance with applicable regulations in non-US jurisdictions.

STOCK RATINGS

Morgan Stanley uses a relative rating system using terms such as Overweight, Equal-weight, Not-Rated or Underweight (see definitions below). Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold and sell. Investors should carefully read the definitions of all ratings used in Morgan Stanley Research. In addition, since Morgan Stanley Research contains more complete information concerning the analyst's views, investors should carefully read Morgan Stanley Research, in its entirety, and not infer the contents from the rating alone. In any case, ratings (or research) should not be used or relied upon as investment advice. An investor's decision to buy or sell a stock should depend on individual circumstances (such as the investor's existing holdings) and other considerations.

Global Stock Ratings Distribution

(as of May 31, 2026)

The Stock Ratings described below apply to Morgan Stanley's Fundamental Equity Research and do not apply to Debt Research produced by the Firm.

For disclosure purposes only (in accordance with FINRA requirements), we include the category headings of Buy, Hold, and Sell alongside our ratings of Overweight, Equal-weight, Not-Rated and Underweight. Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold, and sell but represent recommended relative weightings (see definitions below). To satisfy regulatory requirements, we correspond Overweight, our most positive stock rating, with a buy recommendation; we correspond Equal-weight and Not-Rated to hold and Underweight to sell recommendations, respectively.

Stock Rating Category	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1542	42%	465	51%	30%	707	43%
Equal-weight/Hold	1571	43%	369	40%	23%	723	44%
Not-Rated/Hold	3	0%	0	0%	0%	1	0%
Underweight/Sell	551	15%	86	9%	16%	201	12%
Total	3,667		920			1632	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

Analyst Stock Ratings

Overweight (O). The stock's total return is expected to exceed the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Equal-weight (E). The stock's total return is expected to be in line with the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Not-Rated (NR). Currently the analyst does not have adequate conviction about the stock's total return relative to the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Underweight (U). The stock's total return is expected to be below the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.

Analyst Industry Views

Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.

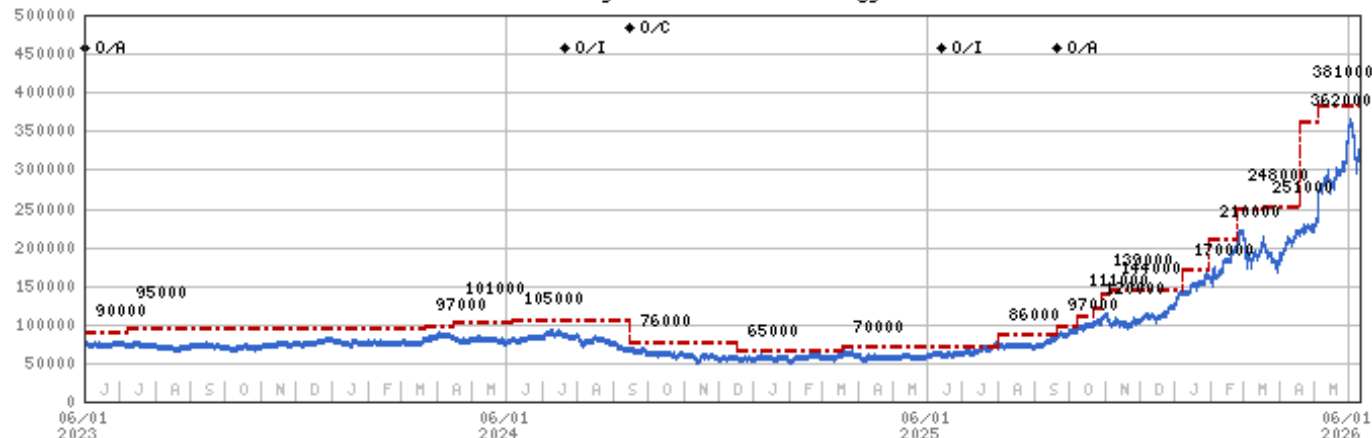
In-Line (I): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be in line with the relevant broad market benchmark, as indicated below.

Cautious (C): The analyst views the performance of his or her industry coverage universe over the next 12-18 months with caution vs. the relevant broad market benchmark, as indicated below.

Benchmarks for each region are as follows: North America - S&P 500; Latin America - relevant MSCI country index or MSCI Latin America Index; Europe - MSCI Europe; Japan - TOPIX; Asia - relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Stock Price, Price Target and Rating History (See Rating Definitions)

Samsung Electronics (005930.KS) - As of 06/10/26 GMT in KRW
Industry : S. Korea Technology



Stock Rating History: 6/1/21 : O/A; 7/19/21 : O/I; 8/12/21 : O/C; 10/4/22 : O/A; 7/21/24 : O/I; 9/15/24 : O/C; 6/13/25 : O/I; 9/21/25 : O/A

Price Target History: 5/18/21 : 93000; 6/8/21 : 98000; 8/12/21 : 89000; 9/15/21 : 95000; 12/3/21 : 97000; 3/18/22 : 95000; 4/28/22 : 85000; 6/10/22 : 80000; 7/5/22 : 75000; 7/22/22 : 70000; 9/17/22 : 68000; 3/21/23 : 70000; 5/30/23 : 90000; 7/7/23 : 95000; 3/22/24 : 97000; 4/16/24 : 101000; 6/6/24 : 105000; 9/15/24 : 76000; 12/18/24 : 65000; 3/19/25 : 70000; 8/1/25 : 86000; 9/21/25 : 97000; 10/8/25 : 111000; 10/23/25 : 120000; 10/30/25 : 139000; 11/5/25 : 144000; 1/8/26 : 170000; 1/30/26 : 210000; 2/24/26 : 248000; 3/18/26 : 251000; 4/19/26 : 362000; 5/6/26 : 381000

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target — No Price Target Assigned (NA)

Stock Price (Not Covered by Current Analyst) — Stock Price (Covered by Current Analyst) —

Stock and Industry Ratings (abbreviations below) appear as ♦ Stock Rating/Industry View

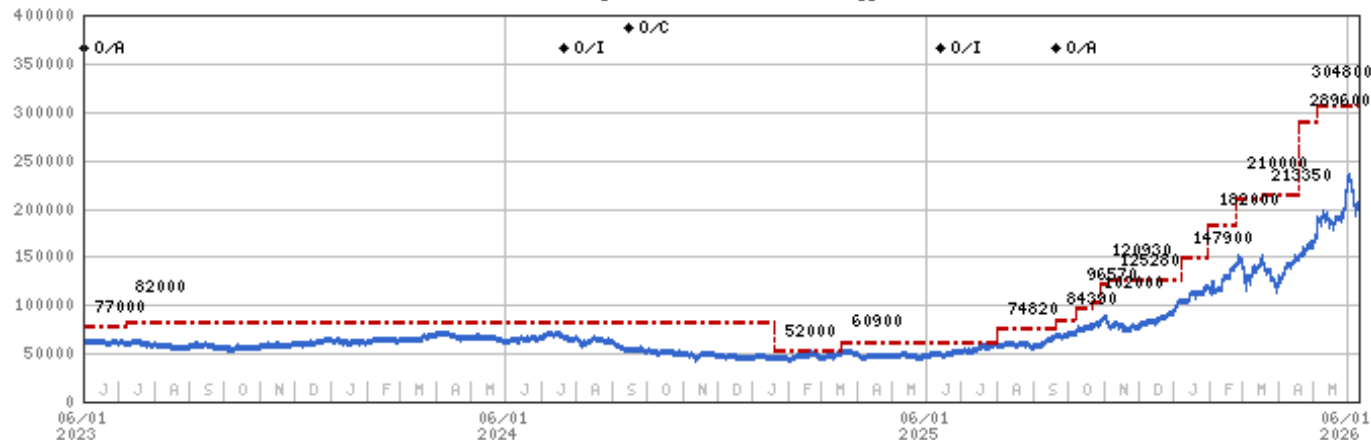
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)

Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Samsung Electronics (005935.KS) - As of 06/10/26 GMT in KRW
Industry : S. Korea Technology



Stock Rating History: 6/1/21 : O/A; 7/19/21 : O/I; 8/12/21 : O/C; 10/4/22 : O/A; 7/21/24 : O/I; 9/15/24 : O/C; 6/13/25 : O/I; 9/21/25 : O/A

Price Target History: 5/18/21 : 80000; 6/8/21 : 84000; 8/12/21 : 77000; 9/15/21 : 82000; 12/3/21 : 83000; 3/18/22 : 82000; 4/28/22 : 73000; 6/10/22 : 69000; 7/5/22 : 65000; 7/22/22 : 60000; 9/17/22 : 58000; 3/21/23 : 60000; 5/30/23 : 77000; 7/7/23 : 82000; 1/20/25 : 52000; 3/19/25 : 60900; 8/1/25 : 74820; 9/21/25 : 84390; 10/8/25 : 96570; 10/23/25 : 102000; 10/30/25 : 120930; 11/5/25 : 125280; 1/8/26 : 147900; 1/30/26 : 182000; 2/24/26 : 210000; 3/18/26 : 213350; 4/19/26 : 289600; 5/6/26 : 304800

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target — No Price Target Assigned (NA)

Stock Price (Not Covered by Current Analyst) — Stock Price (Covered by Current Analyst) —

Stock and Industry Ratings (abbreviations below) appear as ♦ Stock Rating/Industry View

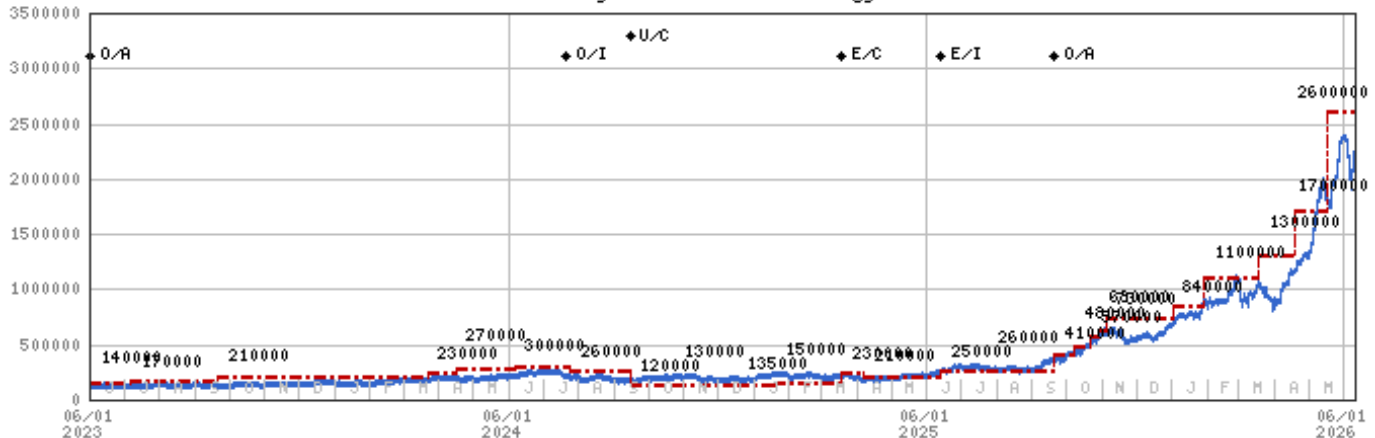
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)

Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

SK hynix (000660.KS) - As of 06/10/26 GMT in KRW
Industry : S. Korea Technology



Stock Rating History: 6/1/21 : O/A; 7/19/21 : O/I; 8/12/21 : U/C; 12/3/21 : E/C; 2/11/22 : O/C; 7/22/22 : E/C; 10/4/22 : O/A; 7/21/24 : O/I; 9/15/24 : U/C; 3/19/25 : E/C; 6/13/25 : E/I; 9/21/25 : O/A

Price Target History: 5/18/21 : 1460000; 6/8/21 : 1560000; 8/12/21 : 800000; 9/15/21 : 880000; 12/3/21 : 1100000; 12/23/21 : 1250000; 1/24/22 : 1300000; 1/28/22 : 1360000; 2/11/22 : 1550000; 3/18/22 : 1500000; 4/27/22 : 1300000; 6/10/22 : 1200000; 7/5/22 : 1100000; 7/22/22 : 1050000; 10/4/22 : 1300000; 12/7/22 : 1200000; 3/21/23 : 1100000; 5/30/23 : 1400000; 7/7/23 : 1700000; 9/21/23 : 2100000; 3/22/24 : 2300000; 4/16/24 : 2700000; 6/6/24 : 3000000; 7/25/24 : 2600000; 9/15/24 : 1200000; 10/24/24 : 1300000; 12/18/24 : 1350000; 1/20/25 : 1500000; 3/19/25 : 2300000; 4/7/25 : 2100000; 6/13/25 : 2500000; 7/24/25 : 2600000; 9/21/25 : 4100000; 10/8/25 : 4800000; 10/23/25 : 5700000; 10/29/25 : 6300000; 11/5/25 : 7300000; 1/2/26 : 8400000; 1/30/26 : 11000000; 3/18/26 : 13000000; 4/19/26 : 17000000; 5/18/26 : 26000000

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target --- No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst) --- Stock Price (Covered by Current Analyst) ---
Stock and Industry Ratings (abbreviations below) appear as ♦ Stock Rating/Industry View
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)
Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

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